

Shahjalal University of Science and Technology
Department of Computer Science and Engineering
2nd Year 1st Semester Final Examination, Jun 2019 (Session: 2017-18)
Course Code: CSE 233 Credits: 3 Course Title: Object-Oriented Programming Language
Time: 3 hrs Total Marks: 100

Group A
[Answer all the questions]

1. Answer any FIVE

5*2=10 n

- a) Determine whether the following statement is true or false and explain your answer:
"Constructors can be either static or non-static."
- b) Can we assign any reference to "this" variable? Explain your answer.
- c) What is the purpose of declaring a variable or method as final?
- d) What are the purposes of super keyword?
- e) What happens when the constructor of a class is declared as "private"?
- f) What is the difference between exception and error in JAVA?
- g) What is the default value of the local variables in JAVA?
- h) Can we change the scope of the overridden method in the subclass?

2. Answer any FOUR

4*5=20

- a) Define True/False. If false give the correct answer.

5*1 m

- i. All exception types are subclasses of the Exception superclass.
- ii. Static nested classes are called Inner class.
- iii. Synchronized methods can only be accessed by one thread at a time.
- iv. String is an immutable primitive data type.
- v. An inner class can access a private member of the outer class.

- b) Determine the following code is correct or incorrect. If the code is correct simply write down the output, if incorrect, specify why and rewrite the correct code.

5 und
stat

```
1. class A {
2.     static void staticMethod() {
3.         System.out.println("Static Method");
4.     }
5. }
6. public class MainClass {
7.     public static void main(String[] args) {
8.         A a = null;
9.         a.staticMethod();
10.    }
11. }
```

- c) Check the following java program carefully, you can insert as many as statements you want but you can't delete or edit any statement from MainClass.

JA
As

```
1. public class MainClass {
2.     public static void main(String[] args) {
3.         MySystem.out.println("Static Method");
4.     }
5. }
```

- d) What will be the output of the following code? Explain.

5 P:

<pre>package pkgA; public class Frog{ int a = 5; protected int b = 6; public int c = 7; } package pkgB; import pkgA.*</pre>	<pre>public class clover{ public static void main(String[] args){ Frog f = new Frog(); System.out.println("a = " + f.a); System.out.println("b = " + f.b); System.out.println("c = " + f.c); }}</pre>
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- e) What are the ways in which you can instantiate a thread? What are the advantages of using multiple threads in a program?

5 N

- f) Write a code using Java threading where a Deadlock condition must occur.

5 N

3. Answer any TWO

2*10

- a) Define the class *Point*, *Shape*, *Rectangle* and *Circle* for the following program. All member variables of these classes should be private.

10

Classes,
Inheritance

```
01 public class ObjectToClass {
02     public static void main (String[] args) {
03         Point p = new Point(1,2);
04         System.out.println("Point:"+p.getX()+" "+p.getY());
05         Rectangle r = new Rectangle(p,20.0,40.0);
06         System.out.println("R:"+r.getHeight()+" "+r.getWidth());
07         System.out.println("R-Area:"+r.area());
08         r.setWidth(8.0); r.setHeight(6.0);
09         Shape s = r; System.out.println("S-Area:"+s.area());
10         System.out.println("S-Point:"+s.getPoint().getX()
11                                 +", "+s.getPoint().getY());
12         p.setX(4); p.setY(6);
13         Circle c = new Circle(p,5.0);
14         System.out.println("C:"+c.getRadius()+" C-Area:"+c.area());
15         c.setRadius(4.0);
16         s = c; System.out.println("S-Area:"+s.area());
17     }
18 }
```

Output of this program:

```
Point:1,2
R:40.0,20.0
R-Area:800.0
S-Area:48.0
S-Point:1,2
C:5.0,C-Area:78.5
S-Area:50.24
```

- b) Once there was a goal keeper in Portugal who used to believe in luck. He threw dice to decide whether he should jump left or right during penalty shootout. If he used to get three or five he used to jump left. Penalty shooters also had two options like the goalkeeper. If both the goal keeper and shooter chose the same side then the goal is saved. If they chose opposite sides then it was celebrated as goal. Now, implement this goalkeeper-penalty shooter problem for five penalties by using multi-threaded programming.

10

Multi Thre
Programm

- c) Check the following java program carefully.

10

Exception
Handling

```
01 class ExceptionExam {
02     public static void m() { throw new NullPointerException("Null"); }
03     public static void m(int e) {
04         try { System.out.println("e = "+e); return;}
05         finally { System.out.println("not-reachable!"); } }
06     public static void main(String exc[]) {
07         int e = exc.length;
08         System.out.println("Total Exceptions: "+e);
09         if (e == 0) e = 10 / e;
10         try {
11             if (e == 1) e = e / (e - e);
12             try {
13                 if (e == 2) { int a[] = {1}; a[1] = 99;
14                     System.out.println("a[1] = " + a[1]);
15                     else if (e == 3) m();
16                     else m(e);
17                 } catch (ArrayIndexOutOfBoundsException ex) {
18                     System.out.println("Divide by 0");
19                 } finally { System.out.println("Un-reachable code"); }
20             } catch (ArithmeticException NullPointerException ex) {
21                 System.out.println("Array index out-of-bounds or Null");
22             } catch (Exception ex) {
23                 System.out.println("Any Exception");
24             } finally { System.out.println("Program Completed"); }
25         } System.out.println("Exam Ended");
26     }
27 }
```

and write the output for the following commands:

- i. java ExceptionExam
- ii. java ExceptionExam e1
- iii. java ExceptionExam e1 e2
- iv. java ExceptionExam e1 e2 e3
- v. java ExceptionExam e1 e2 e3 e4

Group B

[Answer all the questions]

4. Answer any FIVE

5*2

- a) What is the relation between JDK, JRE, and JVM?
- b) Why is it possible for an interface to extend another two interfaces in java?
- c) What gives Java its 'write once and run anywhere' nature? ✓
- d) What happens if a class partially implements an interface? ✓
- e) What happens when an inner class is declared as Final?
- f) What do you think the following fragment of code does? ✓
 1. MyClass c=new MyClass();
 2. MyClass c2=c;
 3. MyClass c3=new MyClass();
- g) Does Java support multiple inheritances? If not why? ✓
- h) What are the restrictions that are applied to the Java static methods? ✓

5. Answer any FOUR

4*5

- a) Define True/False. If false give the correct answer. 5*1
 - i. Inner class cannot be public.
 - ii. Since constructor does not have any return type, it cannot be overloaded.
 - iii. Protected members of a class cannot be accessed from non- subclass within same package.
 - iv. Abstract class cannot have static method.
 - v. Static interface methods are not inherited by either an implementing class or a sub-interface.
- b) Is there any way to throw an exception from a method without adding throws declaration? If possible show an example, if not write the reason. 2+3

- c) Briefly describe the garbage collection mechanism of JAVA. Consider the following code and determine at what point is the Bar object, created on line 6, eligible for garbage collection? 2+3

```
1. class Bar {
2. }
3.
4. public class Test {
5.     Bar doBar() {
6.         Bar b = new Bar();
7.         return b;
8.     }
9.     public static void main (String args[]) {
10.         Test t = new Test();
11.         Bar newBar = t.doBar();
12.         System.out.println("newBar");
13.         newBar = new Bar();
14.         System.out.println("finishing");
15.     }
16. }
```

- d) Can you import same package or class twice? Will the JVM load the package twice at runtime? How many objects will be created in the following code? 3+2

```
String s1="Welcome";
String s2="Welcome";
String s3="Welcome";
```

e) Write down the output of the following program segment.

```
int p[] = {10, 23, 2, 121, 232, 211};
```

```
for(int i=1; i<p.length; i++)
```

```
    if(p[i]%2==0)
```

```
        p[i] = p[i] + p[i-1];
```

```
for (int i=0; i<p.length; i++)
```

```
    System.out.println(int p[i];)
```

f) Show an example of "Array index out of bound exception" by writing a Java Program.

6. Answer any TWO

- a) i) Explain the basic principles of OOP.
ii) What is method overloading and overriding in Java?
iii) Will the following program work properly? Does the program represent polymorphism property of Java and how?

```
public class Main{  
    public static void main(String[] args){  
        main();  
    }  
    Static void main(){  
        System.out.println("It is main method");}}  
}
```

- b) Given an ArrayList with duplicate values, the task is to remove the duplicate values from this ArrayList in Java.

Input: List = [1, 10, 2, 2, 10, 3, 3, 3, 4, 5, 5]
Output: List = [1, 10, 2, 3, 4, 5]

- c) What are the restrictions of generics? How generics improve type safety? Write a generic method reverse Array that reverses the order of elements in a array. Use this generic method in a program to reverse arrays of different types. Print each array both before and after calling reverseArray method.